

Let's Get Our Heads Out of the Clouds

A scalable and sustainable
approach to HPC Training
Labs

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Introduction

The Problem

The training of new and existing HPC (High Performance Computing) practitioners is considered a priority for HPC as a whole.

The Original Solution

Originally, HPC practitioners would be trained face to face from real HPC resources or cloud services.

Modern Solutions

Training material was transferred onto Digital Delivery Platforms, and the use of VMs for hands-on training began to increase. This was done to decrease reliance on existing HPC infrastructure.

Goal of the Paper

This paper aimed to showcase a new solution to offering HPC System Administrator training that is more sustainable, scalable, and can be done offline.

Context

- HPC plays a critical role in fields like AI, climate modeling, and healthcare.
- This increases the demand for trained professionals.
- By offering a cost-effective, replicable training model, the authors hope to support workforce development for data-intensive industries.



Virtual Machines (VMs)

- Virtual Machine (VM) is software-simulated computer environment running off of the resources of a host's computer.
- Virtualization is the process of creating and running multiple VMs on one host system.

Merits of VMs as a Training Resource

Reduces Support Costs

Reduces the amount of time and effort required to help with and deliver the exercise by using mainstream desktop hypervisors and VM deployment tools.

“Golden Image”

Reduces setup costs by preparing a “Golden Image” with all the tools and software required for the course’s needs. This can easily be replicated on demand for participants.

Reducing Actual Costs

Locally hosted VMs can reduce costs for the course provider, as computation is performed on participant’s machines.

Proven Effectiveness

Other professors have found that utilizing VMs is a great way to simulate a real environment at a greater scale for labs.

Classifying the System Administrator Audience

RCE (Resource-Constrained Environment)

RCEs (Resource Constrained Environments) lack access to at least two of the following:

- Access to advanced computing infrastructure (that can be purposed for scientific computing).
- On-Premise/Remote-Access Computing Resources.
- Access to stable and performant internet.
- Reliable Technology Service levels (network connectivity, power supply, etc).

Classifying the System Administrator Audience

Non-RCE

- Non-RCE environments do not face the same issues as RCEs, and are not the target audience for this approach. A traditional approach would fit better.
- However, Non-RCEs are not free of issues (discussed later).



Hosted HPC Resources for Training

Limitations of Traditional HPC Training

Limited Class Size

The Class size may be limited due to the amount of available compute resources.

Dependence on Timed Availability

The availability of training may be reliant on the timing of when compute infrastructure is available for use.

Limited Teaching Resources

Even when resources and compute environments are available for use, it may only be for a limited time.

Inflexibility

The aforementioned limitations limit the time flexibility of training for both the participants and teachers. This can cause the training opportunity to be lost, as the limited resources and availability pass.

Hosted HPC Resources for Training

Limitations of Traditional HPC Training

- Restricted compute resources hinder class sizes and scalability.
- The reliance on timed availability of resources limits accessibility and long term impact of training.
- Cloud Failures could impact training participants.
- No straightforward method for participants to access/reference their original training environment after training. This can stunt information retention or further growth.

Use Case: Training HPC System Admins in RCEs

- As HPC projects expand, brand new deployments necessitate the training of HPC System Administrators in a RCE.
- New HPC sites would not have access to reliable internet connectivity or cloud resources.
- The virtual HPC Training Lab designed in this paper is designed to provide effective, sustainable, and scalable training with these limitations in mind.

The Case for a “Takeaway Lab”

- To address global needs, scalable and sustainable training solutions are needed.
- Trainees worldwide could benefit from resources that persist beyond the course duration.
- Trainees should have the option to retain HPC resources post-training, enabling continued skill development.

The HPC lab model created in this paper emphasizes a self paced, reproducible, and self hosted design emulating a basic cluster.

A Model for HPC Training

- The virtual HPC lab runs on personal computers, requiring no cloud infrastructure or high-end hardware.
- Participants can engage in training at their convenience, accommodating diverse schedules.
- Lessons learned from prior methodologies ensure effectiveness and ease of use.
- The lab is not proposed as the definitive tool for all HPC training but as a demonstration of principles for scalability, which aims to inspire new models for HPC training that adapt to evolving participant needs.

Online Delivery Vs. Online Training

- Online hosting ensures global accessibility and scalability of training materials.
- Training can also be adapted for offline distribution, removing reliance on special infrastructure.
- Blended learning models (combining in-person and online methods) increase accessibility without sacrificing effectiveness.
- Virtual labs allow training to scale to broader audiences without significant changes to existing course content.

Virtual HPC Lab Differences from Traditional HPC Training

1. Content is available in two formats - video tutorials (youtube) and a gitlab interactive guide.
2. The lab uses the OpenHPC software stack to create the HPC environment.
3. *No Cloud* - The HPC components are provisioned by *Hashicorp Vagrant* (development and creation of lightweight and portable development environments), and deployed with Oracle's Virtualbox.
4. VirtualBox is used to emulate a 3 node virtual cluster, which can run on any computer with at least 3-4GB of RAM.
5. The training lab can be modified to be completely done offline (all components are downloadable).
6. Training is entirely self-paced and available on demand, with no cut-off to access HPC Resources.
7. The training lab has fewer restrictions to infrastructure modifications (users can expand or reduce the scale of the lab as they wish).
8. Freedom to fail - the localized lab can use development snapshots to preserve development states in case something goes wrong.
9. The virtual cluster remains on the participant's computer until they choose to remove it.
10. The finished virtual management node can be tweaked to manage physical HPC infrastructure.

Benefits and Downsides

- No need to troubleshoot or worry about remote connections.
- No limit to the amount of participants who can partake in training.
- However, locally hosted VMs can have poor performance, as well as requiring at least some amount of RAM and disk storage space (7GB of ram, 6GB of disk space).
- The Authors have not yet run into a situation where a user could not complete the HPC training due to this.
- If need be, the cluster can be reduce to two nodes, which would only need 4gb of ram and less storage space.

Support Materials



The training lab design prioritizes scalability, inspired by Massive Open Online Courses (MOOCs) and cloud-based labs that minimize technical hurdles while teaching critical cluster computing concepts. The materials support scalable delivery, including offline consumption capabilities.

Support Materials

Virtual Lab Guide



A detailed virtual lab guide was created to help users deploy a 3-node virtual cluster using Vagrant, VirtualBox, and OpenHPC. It includes high-level principles, supplementary insights into the HPC software stack, and is hosted on [GitLab Pages](#) for accessibility.

Support Materials

Youtube Video Guide



Complementary video guides hosted on [Youtube](#) enhance learning, offering offline access* and automatic captions in multiple languages. This setup ensures accessibility for areas with unreliable electricity or network issues.

OpenHPC

OpenHPC is the chosen software stack due to its popularity, ease of implementation, and strong community support. It simplifies deploying traditional HPC software components and is widely adopted in the HPC Ecosystems Community.



No Cloud

The lab uses VirtualBox and Vagrant for local deployment, avoiding issues with remote connectivity, cloud credentials, and provisioning delays. It emulates many of the benefits of traditional hosted delivery platforms while avoiding the problems related to remote/hosted services.

VirtualBox and Vagrant

VirtualBox was chosen for its cross-platform compatibility and wide usage, despite challenges of setup-complexity and lack of user skills (supplemented by the guides and vagrant).

Vagrant was introduced to manage configuration parity, reduce friction, and streamline setup. Vagrant ensures consistency and simplifies the creation, distribution, and updates of virtual environments.

More Benefits of the Training Lab

Potential for Offline Delivery

The lab can operate entirely offline by synchronizing Linux repositories and cloning the GitLab repository locally. This supports participants in regions with limited internet access.

Self-Paced and On Demand

Participants access the lab materials at their convenience, overcoming travel costs, scheduling conflicts, and time zone challenges. The on-demand design ensures accessibility without monetary or logistical constraints.

Fewer Restrictions

Participants can adjust RAM, CPU cores, and other configurations based on their machine's resources. Additional virtual nodes can also be added for further exploration.

Freedom to Fail

Localized labs allow participants to create snapshots, enabling them to save progress and recover from mistakes, encouraging experimentation without fear of permanent loss.

Persistent Images

- The virtual lab remains indefinitely on participants' systems, enabling continued learning or deeper exploration beyond the training period.
- While persistent, participants are encouraged to keep software updated for optimal performance.

Virtual to Physical

Participants can replicate their virtual lab setup on physical systems or link the virtual management node to physical infrastructure with minimal adjustments. This bridges training with real-world application, adding practical value.

Existing Alternatives

- There are many existing HPC conferences (ISC, SC, and PEARC) that offer hands on training opportunities
- However, they are in the northern hemisphere, which causes geographic and time-based accessibility issues for people who are too far away to participate.
- It is intended for “Experienced Linux System Administrators”.
- There has only been 15 hours of collective training from official OpenHPC initiatives.

Table 1: List of Known Formal Training for OpenHPC

Event	Duration in hours	Year & Month	Description
SC20	3	2020-11	Tutorial
SC19	1	2019-11	Birds of a Feather
PEARC19	3.5	2019-07	Tutorial
ISC 2019	1	2019-06	Birds of a Feather
HPCKP'19	0.5	2019-06	Presentation
SC18	1 unknown	2018-11	Birds of a Feather Tutorial
DAAC 2018	unknown	2018-11	Presentation
Open Source Summit 2018	3	2018-08	Tutorial
MVAPICH'18	unknown	2018-08	Presentation
DevConf.CZ 2018	0.25	2018-01	Presentation
SC17	1	2017-11	Birds of a Feather
MVAPICH'17	unknown	2017-08	Presentation
PEARC 17	3.5	2017-07	Tutorial
ISC 2017	1	2017-06	Birds of a Feather
HPCKP 2017	0.5	2017-06	Presentation
SC16	1	2016-11	Birds of a Feather
MVAPICH 2016	unknown	2016-08	Presentation
ISC 2016	1	2016-06	Birds of a Feather
FOSDEM 2016	0.45	2016-01	Presentation

Results and Impact

“Since the launch of the OpenHPC1.3.x Virtual Lab in October 2020, there have been 226 participants trained in six formal online workshops hosted to date, with more than 5,500 views of the accompanying online training videos. Notably, the Virtual HPC Lab has attracted participants from outside of Africa, attesting to the global relevance as on demand virtual OpenHPC Training Lab”.

Evaluation and Assessment (1)

Quantitative Feedback

Regular participant assessment is a key component in enhancing online learning efficacy.

For the Virtual Lab, evaluation can be measured by:

1. The number of virtual clusters successfully opened by a participant.
2. The number of engagements with the Youtube/Gitlab Virtual Lab Guides.

Evaluation and Assessment (2)

Limitations

- The aforementioned metrics are meaningless without additional information (unique visitors, number of attempts and revisits, etc).
- To counteract this, registration forms are incorporated into formal events using this training lab.

Evaluation and Assessment (3)

Qualitative Feedback

- Participants are encouraged to provide regular feedback to enhance the quality of the content being delivered in future iterations.

Risks of Online HPC Training (1)

Risks

1. High Dropout Rates.
2. Isolation (limited peer discussion).
3. Hardware Access (limited hands-on troubleshooting opportunities).

Risks of Online HPC Training (2)

Mitigation Strategies

1. Encouraging Forum Participation.
2. Supplementary Resources (instructional materials).

Risks of Online HPC Training (3)

Technical Challenges

1. Hardware Requirements (minimal risk).
2. Internet Dependency (if not properly set up for offline use).
3. Coordination (For shipping offline HPC lab kits).

Conclusion

- This project expanded the audience reach of the HPC Ecosystems Project, which creator larger and more diverse participation.
- There is no universal solution, so platforms should align with what the audience needs.
- Offline Interaction with a Virtual Lab expands accessibility and offers limitless scalability.

Future Work

- An updated version of the Virtual HPC Lab (OpenHPC 2.x.x) was ready to launch at 2023Q3.
- Further work underway to develop additional modules (standalone or “bolt-on” courses).

THANK YOU

Questions?

References

<https://jocse.org/downloads/jocse-15-1-13.pdf>